

ANDREW LIAO

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EDUCATION

New York University

BA Computer Science and Data Science

New York, NY

2025

Carnegie Mellon University

MS AI Engineering – Research Track

Pittsburgh, PA

Exp. May 2028

- Department Head's Fellowship
- Advisor: Byron Yu
- Coursework: Systems and Toolchains for AI Engineering, Intro to Machine Learning (PhD), Deep Learning Systems

SKILLS

- **Core:** Machine Learning, Deep Learning, Neural Decoding, Statistical Modeling
- **Languages & Libraries:** Python, C++, CUDA, JAX, SQL, PyTorch, scikit-learn, AutoGluon
- **Systems:** Multi-GPU Training, SLURM/HPC, Git, Linux

PUBLICATIONS

- Daniel Doan, **Andrew YS Liao**, Arnav Pallem, Kevin Zhu, Sunishchal Dev, Ashwinee Panda, Shreyas Sunil Kulkarni. *Emergent Misalignment in Mixture-of-Experts Models*. Socially Responsible and Trustworthy Foundation Models Workshop, **NeurIPS 2025**. Also accepted, **AAAI AIGOV 2026**.
- **Y. Liao**, J. Gonzalez, Z. S. Chen. *Characterizing Breathing–Whisking Rhythms and Decoding Whisker Motion Energy*. Poster, **Society for Neuroscience 2025**.
- **Andrew Y.S. Liao**, Orrin Devinsky, Zhe Sage Chen*, Laura Gould*. *Identifying Risk Factors of Prone Position with Sudden Unexpected Deaths in Childhood*. Manuscript in preparation.

RESEARCH EXPERIENCE

Carnegie Mellon University – Yu Group

Graduate Researcher, latents (dimensionality reduction library)

Pittsburgh, PA

Aug 2026 – Present

- Collaborating with advisor Byron Yu and Evren Gokcen on latents, an open-source dimensionality reduction library for latent variable models of neural population activity.

New York University – GRAIL Lab

Undergraduate Researcher, Robot Utility Models

New York, NY

Jan 2025 – Jul 2025

- Trained visuomotor policies on NYU Greene HPC (multi-GPU), achieving robust zero-shot place-and-drop generalization across arbitrary graspable objects with no object-specific fine-tuning.
- Owned end-to-end development on a Hello Robot mobile manipulator: 3D-printed a custom gripper, curated 1,000+ demonstration videos, and diagnosed/resolved sim-to-real failure modes via targeted data augmentation and retraining.

NYU Langone Health

CN3 Lab – Undergraduate Researcher, Multi-Camera Behavioral Pipeline and Neural Decoding

New York, NY

Feb 2025 – Present

- Produced brain-wide R^2 decoding maps (peak $R^2 = 0.49$) from 240,000-sample sessions; first-author abstract accepted at Society for Neuroscience 2025, and the project was accepted by the International Brain Laboratory for further development.
- Cut per-session memory from 50+ GB to 10 GB and achieved 12–30x speedups via mmap-based lag construction and chunked ridge solvers ($X^T X$ accumulation), scaling to 4,500-feature models.
- Built a memory-optimized pipeline synchronizing 80+ multi-camera recordings across 12 labs (100+ animals), extracting aligned behavioral features (nose PCA, whisker energy, pupil tracking, tongue detection).

SUDCRRRC – Undergraduate Researcher, SUDC Risk Modeling

Feb 2025 – Present

- Led ML analysis of a 346-case clinical registry for the Sudden Unexpected Death in Childhood Research and Registry Collaborative: FAMD/MCA dimensionality reduction, UMAP + K-Means clustering (bootstrap ARI = 0.95), and CatBoost ensembles (AUC = 0.78). First-author manuscript in preparation.

Algoverse

Researcher, Emergent Misalignment in Mixture-of-Experts Models

New York, NY / Remote

Jul 2025 – Oct 2025

- Co-first author, NeurIPS FM 2025 / AAAI AIGOV 2026: ran a 30-run 4-bit QLoRA fine-tuning grid across Mixtral-8x7B, Qwen3-30B, and GPT-oss-20B, uncovering a negative correlation between expert sparsity and emergent misalignment, validated via LLM-as-a-judge and StrongREJECT.

Research Mentor, Mechanistic Interpretability

Jun 2026 – Sep 2026

- Mentor 3 teams of undergraduate researchers on mechanistic interpretability, latent reasoning, and alignment projects, running weekly check-ins to scope problems and review experimental design, code, and written work toward publishable contributions.